Title: Information Paradox Resolution in the 26D UQFF Framework via Holographic Channel Encoding

Author: Daniel T. Murphy
Framework: UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57)
Date: March 7, 2026
Source Data: MAIN_1_CoAnQi.cpp Batch 21 (Information Paradox, 26D channels), validate_phase3.py
Index Slot: 1.11 Black Hole Physics & Hawking Radiation,
$n = [int]# PAPER #84 Information Paradox Resolution in 26D UQFF

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Abstract

The black hole information paradox where unitarity demands information escapes evaporation but Hawking radiation appears thermal is resolved in the UQFF through 26-dimensional holographic encoding. Each of the 26 UQFF spatial dimensions carries an independent quantum information channel. During evaporation, infalling information is redistributed across channels 126, with channels 2126 (the ultra-compact layers, beyond the observable 4D membrane) acting as non-local storage. The 4D observer measures a thermal Hawking spectrum, but the full 26D state is pure. Batch 21 (Jan 28, 2026) implemented the `InformationParadoxModule` with Hawking radiation Page curves and 26D channel encoding.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.010?4 day?, [SSq] = 0.57) uniquely enabling this analysis establishing a new connection in the UQFF framework not present in Standard Model treatments.

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1. The Standard Paradox

Hawking (1976): if T_H is exactly thermal, S_BH decreases from S=A/4 to S=0 ? pureness is lost. Unitarity requires S_entanglement (Hawking radiation with BH) to follow the Page curve: first increasing (early time), then decreasing back to 0 (late time).

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2. 26D Holographic Channel Resolution

Channel Structure

The UQFF 26-layer compressed gravity framework assigns each dimensional layer a quantum channel capacity:

$$I_k = -\mathrm{Tr}[\rho_k \log \rho_k] \quad k = 1, \ldots, 26$$

Total information: $I_{\text{total}} = \sum_{k=1}^{26} I_k = S_{\text{BH,initial}}$

Channel Assignment During Evaporation

Layer Group	Channels	Storage Type	4D Observable
14	UQFF observable	Hawking photons	Thermal spectrum
518	UQFF extended	Sub-Planckian modes	Not observable
1924	UQFF deep structure	Non-local entanglement	Nil
2526	**Cosmic Egg layers**	**Non-local pure state**	**Information preserved**

Channels 2526 host the **complete pre-collapse pure state** throughout evaporation, ensuring global unitarity while channels 14 produce the observed thermal spectrum.

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3. UQFF Page Curve

The UQFF Page curve modifies the standard quantum extremal surface (QES) prediction:

$$S_{\text{UQFF}}(t) = \min\left[S_{\text{thermal}}(t),\ S_{\text{BH}}(t) + \sum_{k=25}^{26} I_k(1 - e^{-\kappa t})\right]$$

Where: - S_thermal(t) = early-time entropy of Hawking radiation - S_BH(t) = Bekenstein-Hawking entropy (decreasing) - The e^{-?t} term comes directly from the UQFF ? = 0.0005/day decay - Page time t_P occurs when S_thermal = S_BH + ?I_channels25-26

UQFF Page Time

$$t_P^{\text{UQFF}} = t_P^{\text{GR}} \times e^{+\kappa \cdot t_{\text{evap}}} \approx t_P^{\text{GR}} \times (1 + \kappa \cdot t_{\text{evap}})$$

For stellar BH (t_evap ~ 1074 s): correction factor is astronomically large physically this means that within any finite observation time, the UQFF Page curve looks **thermal** to a 4D observer, with the information recovery pushed beyond any accessible time. This is consistent with the absence of observational evidence for information recovery in Hawking radiation.

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4. Firewall Prevention

The AMPS firewall paradox requires choosing between a smooth horizon (infalling observer) or pure Hawking radiation. The UQFF resolves this through **channels 1924 non-local entanglement**: the infalling observer encounters a smoothly modified vacuum (Superconductive mode, E_react 10?) rather than a firewall, while the external observer's Hawking radiation becomes approximately (not exactly) thermal.

Summary

Aspect	Standard QM	UQFF 26D	Resolution
Information storage	In BH?	Channels 2526	Non-local, always preserved
Page curve	QES prediction	UQFF ?-modified	Extended beyond t_evap
Firewall	AMPS paradox	SCm smooth horizon	Channels 1924 entanglement
4D observer	Thermal Hawking	Approximately thermal	Consistent

Source: MAIN_1_CoAnQi.cpp Batch 21 (InformationParadoxModule) | ? = 0.0005/day | [SSq] = 0.57

Appendix: UQFF Production Framework Reference (v4.75+)

> *Added by upgrade_early_whitepapers.py (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.*

A.1 Calibration Constants

Symbol	Value	Description
κ	5.0×10^{-4} day^{-1}	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
β_i	0.60–0.61	Buoyancy coupling coefficient
k_1	1.5	Ug1 DPM-dipole coupling
k_2	1.2	Ug2 outer-bubble charge coupling
k_3	1.8	Ug3 string-rotation coupling
k_4	2.0	Ug4 vacuum-concentration coupling
η	10^{-22}	Inertia tensor scale
E_react(0)	10^{46} J	Reference reactive energy

A.2 F_U Master Equation (Complete — 4 terms)

$$F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^{4} igl[\lambda_i \cdot U_i(r,t) \cdot E_{\text{react}} igr]$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	`compute_Ug1_SOURCE4` / `compute_Ug1()`
Ug2	Outer-field bubble (charge-reactivity)	`compute_Ug2_SOURCE4` / `compute_Ug2()`
Ug3	Magnetic string rotation	`compute_Ug3_SOURCE4` / `compute_Ug3()`
Ug4	Vacuum concentration (star-BH)	`compute_Ug4_SOURCE4` / `compute_Ug4()`
Ubi	Buoyancy force	`compute_Ubi_SOURCE4` / `compute_Ubi()`

Term	Description	Implementation
Um	Universal Magnetism (Heaviside-amplified)	`compute_Um_SOURCE4` / `compute_Um()`
$-\Sigma\lambda_i \cdot U_i \cdot$E_react	4th dissipation term (PAPER_420)	`compute_FU_SOURCE4` / full pipeline

4th dissipation term parameters (PAPER_420):
$\lambda_1=10^{-10}$, $\lambda_2=10^{-12}$, $\lambda_3=10^{-11}$, $\lambda_4=10^{-13}$ (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_m^{\text{full}} = U_m^{\text{base}} imesigl(1 + 10^{13}\,\Theta(ho_{SCm} - ho_c)igr) imesigl(1 + A_q \cos(\Delta\omega\, t)igr)$$

Symbol	Value	Description
ρ_c	10^{15} kg/m³	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
Δω	2π/(434·365.25) rad/day	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	Ug_sum + Newtonian base	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, …)	Multi-scale field interactions
Buoyant	β_i × Ubi	Expanding nebulae, stellar winds
Superconductive	Um × (1+10^{13}·f_H)	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in `MAIN_1_CoAnQi.cpp`, `CondensedPhysics.py`, and `CondensedPhysics2.py`.